Figure 4.2: Architecture Work Flow (Different Layers)

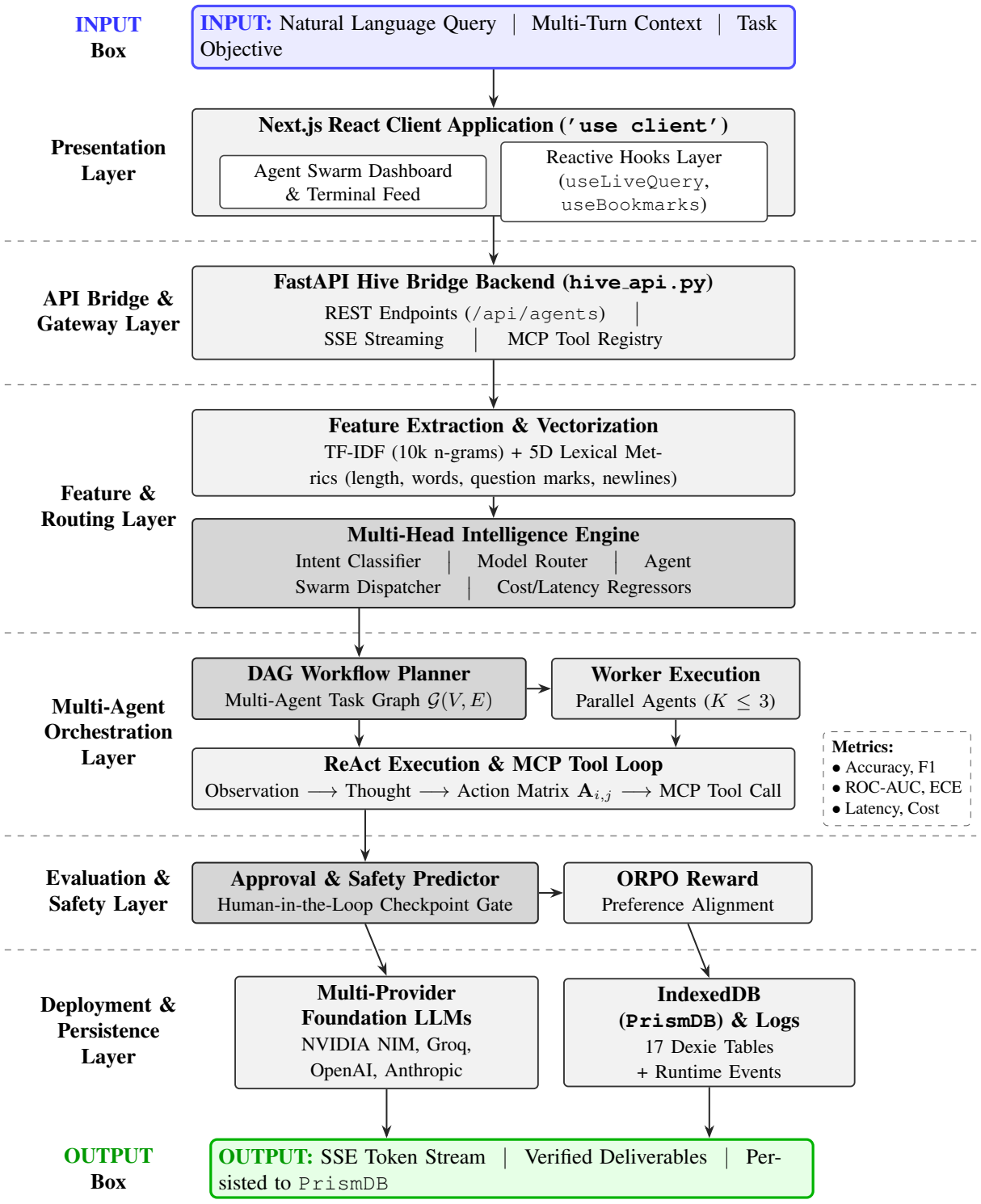


Figure 4.2: Different Layers.

Glossary of Terms & Abbreviations:

- **FastAPI**: High-performance asynchronous Python web framework for backend APIs.
- **REST**: Representational State Transfer (standard HTTP API design pattern).
- **SSE**: Server-Sent Events (persistent HTTP connection for real-time text streaming).
- **MCP**: Model Context Protocol (open standard for agent tool and resource discovery).
- **DAG**: Directed Acyclic Graph (execution plan with task dependency order, no cycles).
- **ReAct**: Reason + Act (agent loop alternating between reasoning and tool calls).
- **ORPO**: Odds Ratio Preference Optimization (direct model alignment without reward model).
- **IndexedDB**: Transactional browser database for client-side persistence (`PrismDB`).
- **Dexie.js / `useLiveQuery`**: Reactive JavaScript engine updating UI on DB changes.
- **ROC-AUC / ECE**: Classification discrimination metric / Expected Calibration Error.